Rodolfo Ostilla Mónico

Associate Professor (Profesor Titular), Fluid Mechanics

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Summary

Fluid dynamicist specializing in the direct numerical simulation of turbulent flows.

Education

Doctor of Philosophy (PhD) — **Twente University**, Physics of Fluids — Enschede, NL — May 2011 – July 2015
- Thesis: Highly turbulent Taylor-Couette flow: direct numerical simulations
- Advisors: D. Lohse (director), R. Verzicco (co-director)

Aerospace Dynamics MSc — **Cranfield University**, Aerodynamics — Cranfield, UK — Sept 2009 – Sept 2010
- Thesis: Aerodynamic flow control through weak plasma interaction

Ingeniero Aeronáutico (5-year degree) — **Universidad de Sevilla**, Aerospace Engineer — Sevilla, Spain — Sept 2005 – Sept 2010

Grado en Filosofía (Philosophy BSc) — **Universidad Nacional de Educación a Distancia (UNED)**, Philosophy — Spain — Oct 2015 – June 2022

Experience

Universidad de Cádiz, Assistant Professor / Profesor Titular — Puerto Real, Spain — July 2026 – present — 4 months
- Department of Mechanical Engineering and Industrial Design, School of Engineering

Universidad de Cádiz, Emergia Postdoctoral Fellow — Puerto Real, Spain — Mar 2022 – July 2026 — 4 years 5 months
- Department of Mechanical Engineering and Industrial Design, School of Engineering

Dept. of Mechanical Engineering, University of Houston, Assistant Professor — Houston, USA — Oct 2017 – Feb 2022 — 4 years 5 months
- Taught undergraduate and graduate courses in fluid mechanics and non-linear dynamics
- Supervised 5 MSc theses, 4 undergraduate research projects, and 1 PhD thesis

Harvard University, Postdoctoral Researcher — Cambridge, MA, USA; Oct 2015 – Sept 2017; 2 years

European Space Astronomy Centre (ESAC), Trainee — Madrid, Spain; Oct 2010 – Mar 2011; 6 months

Departamento de Mecánica de Fluidos, Universidad de Sevilla, Research Assistant — Seville, Spain; Jan 2008 – Jan 2009; 1 year 1 month

Projects

Aeroacústica e hidrodinámica de los instrumentos musicales tipo chimenea — Sept 2023 – Aug 2027
University of Cádiz, competitive research project funded by the Spanish State Research Agency (Proyectos de Generación de Conocimiento 2022). Main PI.
- Total funding: €81,250

CDS&E: Multi-scale Coherent Structure Extraction and Tracking for Modern CFD Data Analysis — Sept 2021 – Aug 2024
University of Houston, competitive research project funded by the U.S. National Science Foundation. Co-PI with Guoning Chen and Di Yang.
- Total funding: $453,000

Particulate Rayleigh-Bénard Flow — Sept 2021 – Aug 2024
University of Houston, competitive research project funded by the U.S. National Science Foundation. Principal Investigator.
- Total funding: $209,666

Controlling Secondary Flows by the Use of Non-Wetting Surfaces — Sept 2019 – Aug 2022
University of Houston, competitive research project funded by the U.S. National Science Foundation. Co-PI with Kamran Alba.
- Total funding: $340,000

Awards

Visions of Fluid Dynamics award (European Fluid Dynamics Conference) — Jan 2025
With W. Shen and X. Zhu, for the video “Vortices vs. Magnetic Fields: Competing Orders in Flux Tubes”.

Emergia Fellowship — Jan 2022
Fellowship funding return to Spain and research at the Universidad de Cádiz (€85,000).

College Junior Teaching Award — Jan 2021

College Junior Research Award — Jan 2021

Vizapalooza competition winner — Jan 2018

Gallery of Fluid Motion award — Jan 2017
For the video “The Emergence of Small Scales in Vortex Ring Collisions”.

ERCOFTAC Da Vinci Award finalist — Jan 2015

Wim Nieuwpoort Award for best national scientific code — Jan 2012

Best MSc thesis (departmental prize) Jan 2010

Royal Aeronautical Society Aerospace Challenge, winning team Jan 2010

Teaching

- Universidad de Cádiz: Mecánica de Fluidos II (Grado en Ingeniería Aeroespacial, 4.5 credits), 2022/23 to 2025/26. Course redesigned by me; covers potential flow, boundary layers, compressible flows, turbulence and CFD labs.
- Universidad de Cádiz: Instalaciones y Máquinas Hidráulicas y Térmicas (Máster Universitario en Ingeniería Industrial, 5 ECTS), hydraulic part, 2026/27 (current).
- Universidad de Cádiz: short course "An Invitation to Computational Fluid Dynamics from Fundamentals to Applications" (20 hours), September 2026.
- University of Houston: Introduction to Fluid Mechanics (2018 to 2021), Fluid Mechanics II / Fluids II (2018, 2019), and Advanced CFD (2019, graduate course developed by me).
- University of Houston: Scientific Computing and Non-linear Dynamics and Chaos (combined undergraduate/graduate, 2020).

Supervision

- Doctoral thesis: V. Jeganathan, “Controlling Secondary Flows in Turbulent Taylor-Couette Flow Using Surface Heterogeneity” (University of Houston, 2022).
- Master theses (University of Houston): S. Soriano (2022), D. Putt (2021), A. Janjua (2020), H. Mitra (2020), A. Mishra (2019).
- Undergraduate theses (TFG), Universidad de Cádiz: main director of 12 completed projects (2025 to 2026) on ventilation, vortex dynamics, CFD of rotors, wings and tsunami flows, and indoor air quality; co-director of 2 more.
- Undergraduate research fellows (University of Houston): N. Betancourt, S. Soriano, R. Bhattarai, C. Tolat.

Service

- Chair (2025), Vice-Chair, and member (2022 to 2025) of the Diversity and Inclusion Committee, APS Division of Fluid Dynamics.
- Session organizer and chair at APS-DFD, Euromech and Spanish Fluid Mechanics Conference meetings; scientific committee of the 22nd International Couette-Taylor Workshop (2023).
- Reviewer for over 100 papers in leading fluid mechanics and physics journals; proposal reviewer for the NSF (US), the European Research Council and Fondecyt (Chile); judge of the Gallery of Fluid Motion (2018, 2021).
- Faculty advisor of the aeromodelling student association UCa&Air, Universidad de Cádiz.

Publications

Elastic turbulence in space
Rodolfo Ostilla-Mónico
doi.org/10.1017/jfm.2024.1214

Fluid dynamical pathways of airborne transmission while waiting in a line
Ruixi Lou, Milo Van Mooy, Gabriel A. Tarditti, Rodolfo Ostilla-Mónico, Varghese Mathai
doi.org/10.1126/sciadv.adw0985

Transition between boundary-limited scaling and mixing-length scaling of turbulent transport in internally heated convection
Sina Kazemi, Rodolfo Ostilla-Mónico, David Goluskin
doi.org/10.1103/PhysRevLett.129.024501

Controlling secondary flows in Taylor-Couette flow using axially spaced superhydrophobic surfaces
Vignesh Jeganathan, Tala Shannak, Kamran Alba, Rodolfo Ostilla-Mónico
doi.org/10.1017/jfm.2023.606

Instability and disintegration of vortex rings during head-on collisions and wall interactions
Aakash Mishra, Alain Pumir, Rodolfo Ostilla-Mónico
doi.org/10.1103/PhysRevFluids.6.104702

Turbulence generation through an iterative cascade of the elliptical instability
Ryan McKeown, Rodolfo Ostilla-Mónico, Alain Pumir, Michael P. Brenner, Shmuel M. Rubinstein
doi.org/10.1126/sciadv.aaz2717

Physical and geometric constraints shape the labyrinth-like nasal cavity
David Zwicker, Rodolfo Ostilla-Mónico, Daniel E. Lieberman, Michael P. Brenner
doi.org/10.1073/pnas.1714795115

Logarithmic mean temperature profiles and their connection to plume emissions in turbulent Rayleigh-Bénard convection
Erwin P. van der Poel, Rodolfo Ostilla-Mónico, Roberto Verzicco, Siegfried Grossmann, Detlef Lohse
doi.org/10.1103/PhysRevLett.115.154501

A pencil distributed finite difference code for strongly turbulent wall-bounded flows
Erwin P. van der Poel, Rodolfo Ostilla-Mónico, John Donners, Roberto Verzicco
doi.org/10.1016/j.compfluid.2015.04.007

Optimal Taylor-Couette flow: direct numerical simulations
Rodolfo Ostilla, Richard J. A. M. Stevens, Siegfried Grossmann, Roberto Verzicco, Detlef Lohse
doi.org/10.1017/jfm.2012.596

Skills

Computational Fluid Dynamics

Applied Fluid Mechanics

Interests

Philosophy of Science

Languages

Spanish
Native speaker

English
Fluent

Mandarin Chinese
B1